Ohm's law, series and parallel circuits

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Find the current.
- $I = 5 \div 10 = 0.5$ A

1b) Find the resistance R.
- $R = 5 \div 0.2 = 25\ \Omega$

1c) Find the battery voltage.
- $V = 0.3 \times 20 = 6$ V

1d) Find the battery voltage (30 mA through 1.5 kΩ).
- $V = 0.03 \times 1500 = 45$ V

2 For each series circuit, calculate the total resistance, the current and the voltage across each resistor.
- a) $R_T = 1520\ \Omega$; $I = 7.89$ mA; $V_1 = 0.16$ V; $V_2 = 11.8$ V
- b) $R_T = 1400\ \Omega$; $I = 8.57$ mA; $V_1 = 0.86$ V; $V_2 = 11.1$ V
- c) $R_T = 1.26$ MΩ; $I = 26.2$ µA; $V_1 = 6.8$ V; $V_2 = 26.2$ V

3 For each parallel circuit, calculate the total resistance, the circuit current and the current in each resistor.
- a) $R_T = 4.8\ \Omega$; $I_T = 2.5$ A; 1 A and 1.5 A
- b) $R_T = 88.6\ \Omega$; $I_T = 1.24$ A; 0.019 A and 1.22 A

4a) For each, calculate the total resistance and the circuit current.
- c) $R_{parallel} = 42.3\ \Omega$; $R_T = 1042\ \Omega$; $I = 0.221$ A
- d) $R_{parallel} = 26.8\ \Omega$; $R_T = 2227\ \Omega$; $I = 0.103$ A

4b) For each, calculate the voltage across the parallel section and the current in each parallel resistor.
- c) $V_p = 230 - 0.221 \times 1000 = 9.3$ V; 0.020 A, 0.014 A, 0.19 A
- d) $V_p = 230 - 0.103 \times 2200 = 2.8$ V; 0.0063 A, 0.0045 A, 0.092 A

5 15 kΩ and 25 kΩ resistors are connected in parallel to a 6 V supply. Calculate the total resistance and the circuit current.
- $R_T = (15 \times 25) \div 40 = 9.38$ kΩ
- $I = 6 \div 9375 = 0.64$ mA

6 2 kΩ, 5 kΩ and 10 kΩ resistors are connected in parallel to a 10 V supply. Calculate the total resistance, the circuit current and the current through the 2 kΩ resistor.
- $1/R_T = 1/2000 + 1/5000 + 1/10\,000$, so $R_T = 1250\ \Omega$
- $I = 10 \div 1250 = 8$ mA
- $I_{2k} = 10 \div 2000 = 5$ mA

7 A burglar alarm has two buzzers (200 Ω and 300 Ω) in parallel on an 8 V supply. Calculate the circuit resistance, the circuit current and the current through the 200 Ω buzzer.
- $R_T = (200 \times 300) \div 500 = 120\ \Omega$
- $I = 8 \div 120 = 0.067$ A
- $I_{200} = 8 \div 200 = 0.04$ A

8 A student needs 50 Ω but only has two 100 Ω resistors. Draw a solution and prove it with a calculation.

- Two 100 Ω resistors in parallel
- $R_T = (100 \times 100) \div 200 = 50\ \Omega$

9 A 6 V supply feeds 30 Ω and 20 Ω resistors in parallel, in series with a 10 Ω resistor. Calculate the total resistance, the circuit current, the voltage across the 10 Ω resistor and the current through the 30 Ω resistor.

- $R_{parallel} = 12\ \Omega$; $R_T = 22\ \Omega$
- $I = 6 \div 22 = 0.273$ A
- $V_{10} = 0.273 \times 10 = 2.73$ V
- $V_{parallel} = 6 - 2.73 = 3.27$ V; $I_{30} = 3.27 \div 30 = 0.109$ A

10a) Calculate the total resistance of the circuit.

- Lower branch: 1200 + 270 = 1470 Ω
- $R_T = (390 \times 1470) \div (390 + 1470) = 308\ \Omega$

10b) The reading on A1 is 0.031 A. Calculate the voltage across the 390 Ω resistor.

- $V = 0.031 \times 390 = 12.1$ V

10c) Calculate the current A2.

- A2 = 12.1 ÷ 1470 = 0.0082 A

10d) Calculate the current A3.

- A3 = 0.031 + 0.0082 = 0.039 A